BANGLA CYBERBULLYING DETECTION USING TRANSFORMER MODELS AND ENSEMBLE LEARNING

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INTRODUCTION

- Social media has greatly increased online communication.
- However, cyberbullying and abusive language are also increasing.
- In Bangla platforms, abuse appears in different forms.
- Manual moderation is difficult due to large data volume.
- NLP-based systems can automatically detect cyberbullying.

PROBLEM STATEMENT

- Most cyberbullying detection systems use binary classification.
- They only identify bully vs non-bully content.
- Different types of cyberbullying are not properly categorized.
- Bangla cyberbullying datasets are limited and imbalanced.
- A more accurate multi-class detection system is needed.

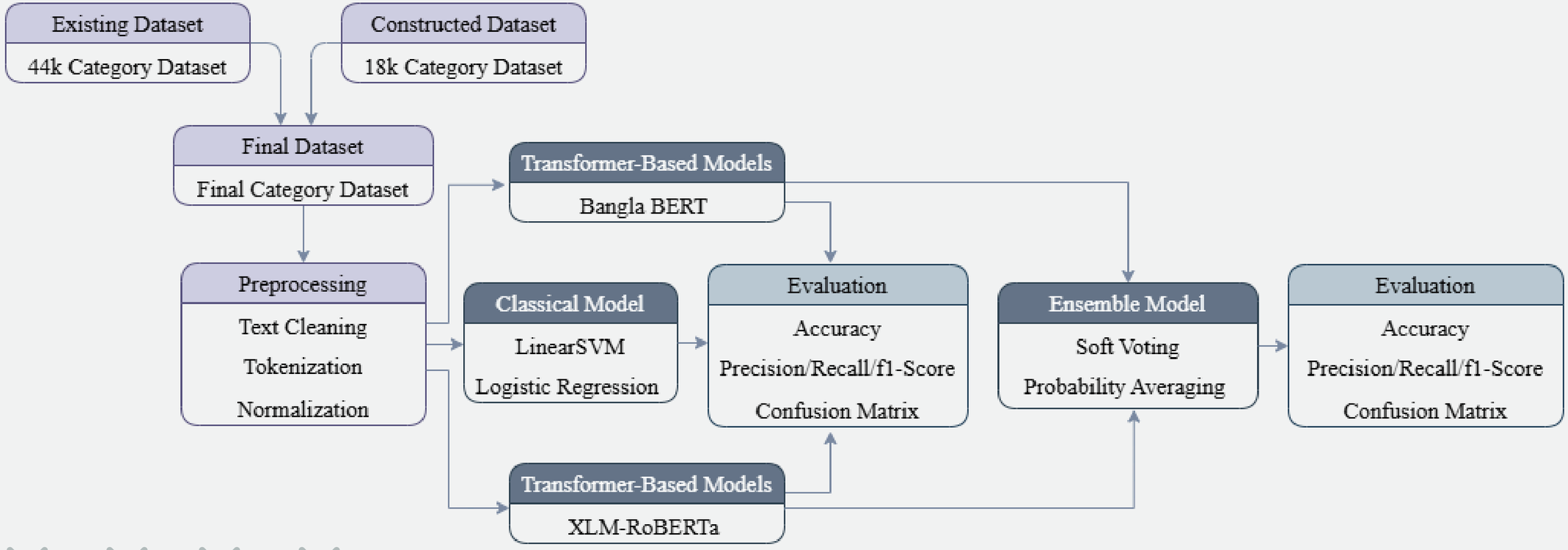
RESEARCH OBJECTIVES

- Build a large Bangla cyberbullying dataset.
- Detect multiple types of cyberbullying.
- Compare classical ML and transformer models.
- Improve performance using ensemble learning.

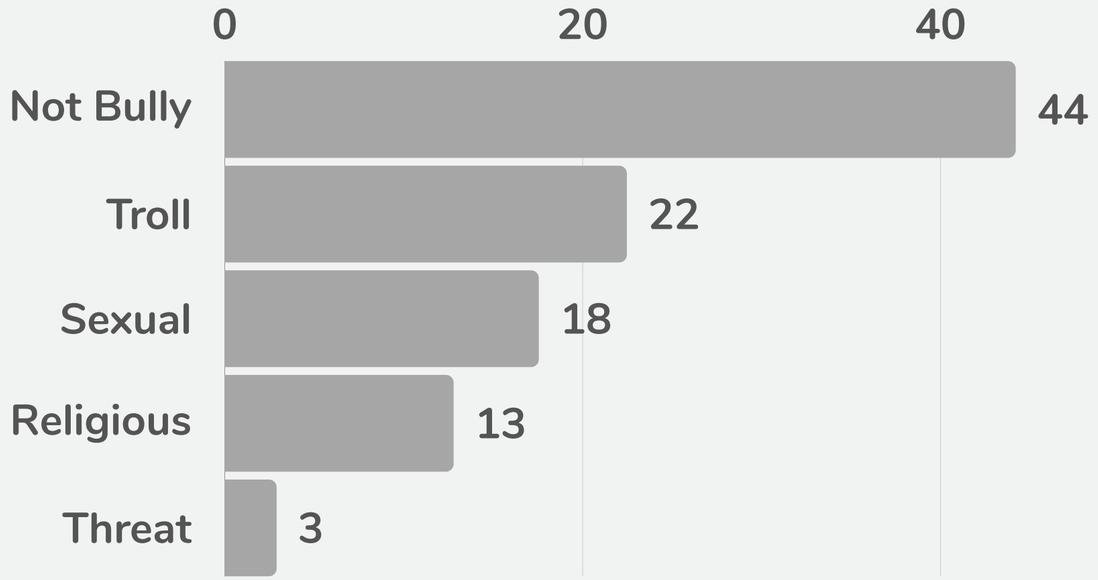
LITERATURE COMPARISON

Study	Model Type	Accuracy
Ahmed et al. (2021)	Deep neural network	87.91%
Wahid and Al Imran (2023)	Multi-feature transformer	86.30%
Mahmud et al. (2023)	Hybrid deep learning models	82.00%
Nath et al. (2024)	Bi-LSTM based deep learning	95.08%
Akhter et al. (2023)	Hybrid machine learning model	98.57%
Proposed Work	Bangla BERT + XLM-R Ensemble	92.20%

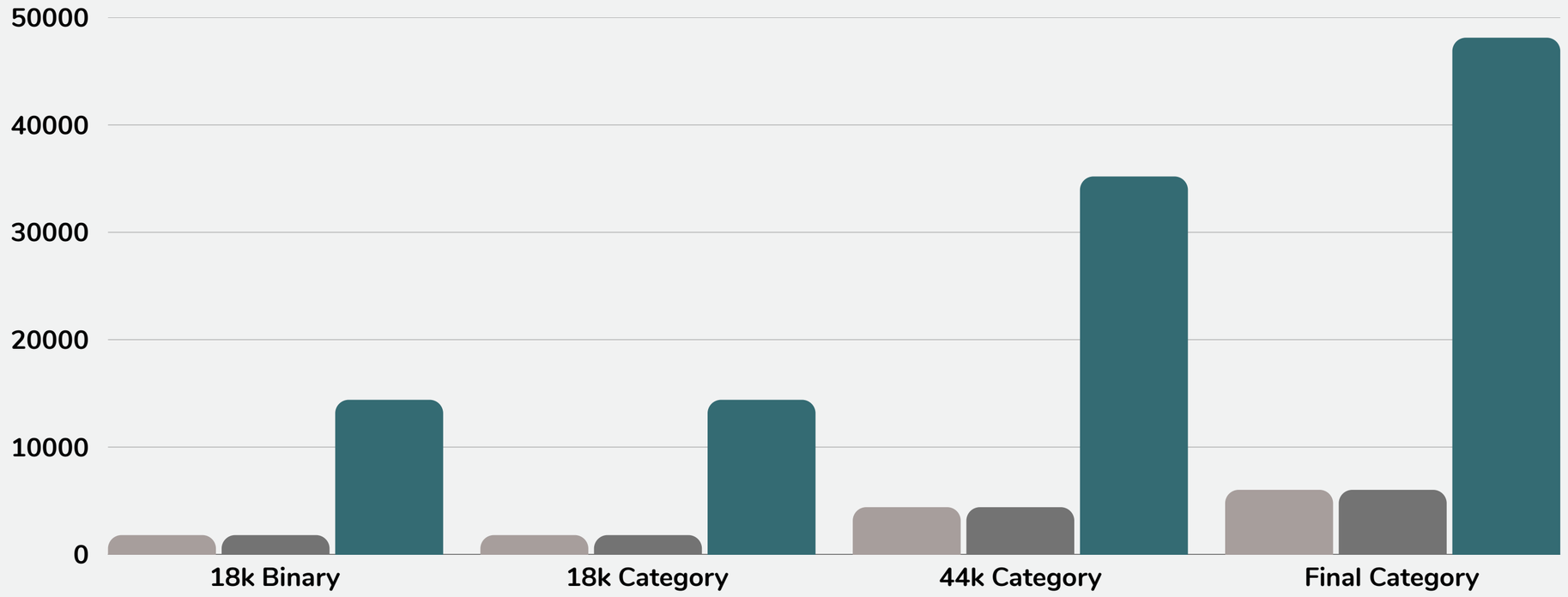
SYSTEM ARCHITECTURE



CLASS DISTRIBUTION



DATASET STATISTICS



MODELS USED

Classical Machine Learning	• Support Vector Machine • Logistic Regression
Transformer-Based	• Bangla BERT • XLM-RoBERTa
Ensemble Model	• Combination of BERT and XLM-R • Soft Voting technique

CLASSICAL MODEL PARAMETERS

Feature Extraction

- Method: TF-IDF
- N-gram range: (1,2)

Binary Classification

- Model: Support Vector Machine
- Class weight: Balanced

Multi-Class Classification

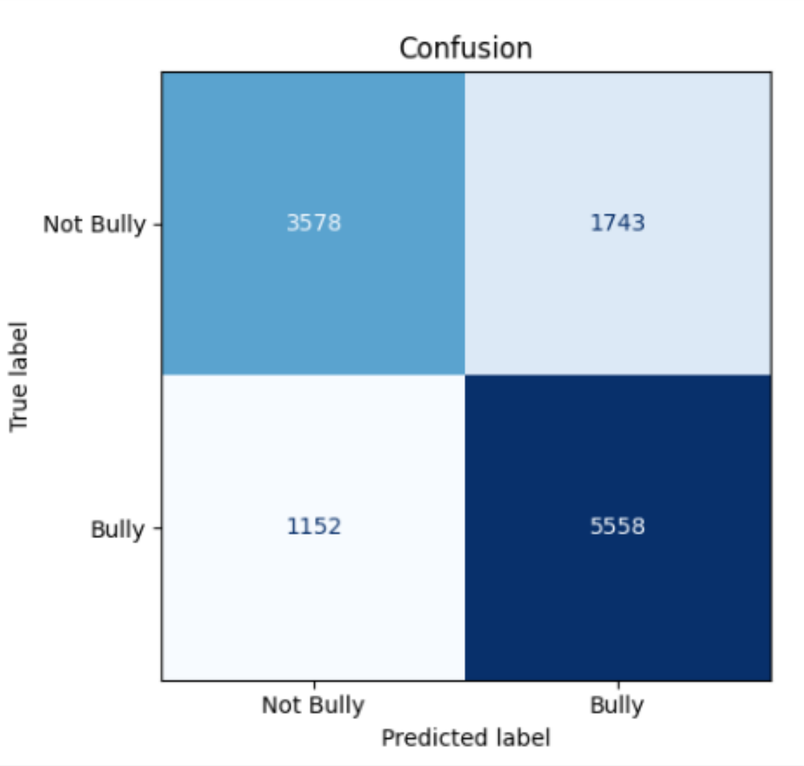
- Model: Logistic Regression
- Class weight: Balanced

CLASSICAL MODEL RESULTS

Binary Classification (Support Vector Machine)

Tag	Precision	Recall	F1-score	Support
Not Bully	0.76	0.67	0.71	5321
Bully	0.76	0.83	0.79	6710
Accuracy			0.76	12031
Macro avg	0.76	0.75	0.75	12031
Weighted avg	0.76	0.76	0.76	12031

CONFUSION MATRIX

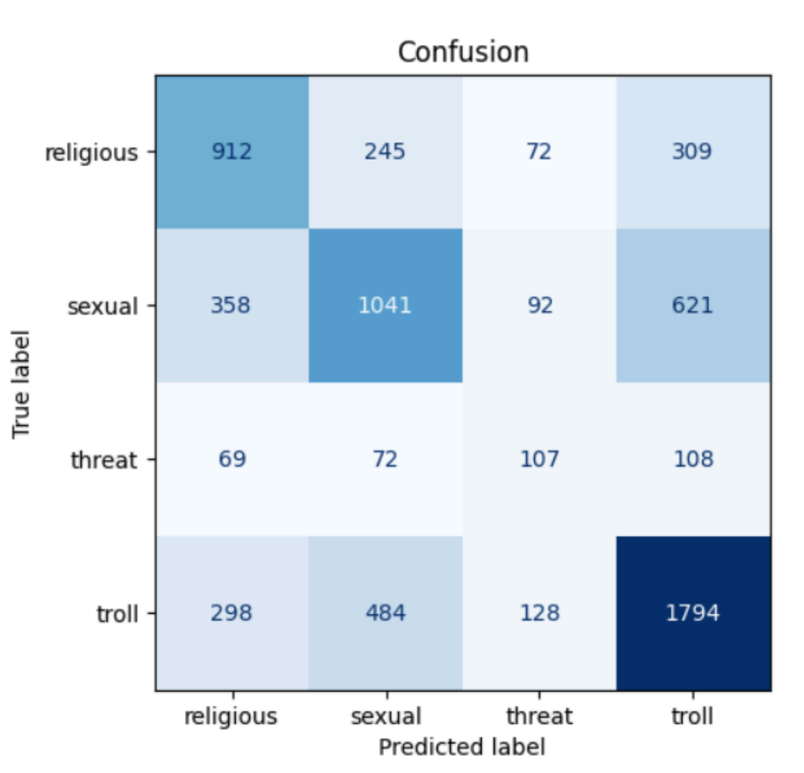


CLASSICAL MODEL RESULTS

Multi-Class Classification (Logistic Regression)

Tag	Precision	Recall	F1-score	Support
Religious	0.56	0.59	0.57	1538
Sexual	0.57	0.49	0.53	2112
Threat	0.27	0.3	0.28	356
Troll	0.63	0.66	0.65	2704
Accuracy			0.57	6710
Macro avg	0.51	0.51	0.51	6710
Weighted avg	0.58	0.57	0.57	6710

CONFUSION MATRIX



BERT MODEL PARAMETERS

Training Setup

- Learning rate: 1.5e-5
- Batch size: 8
- Epochs: 4

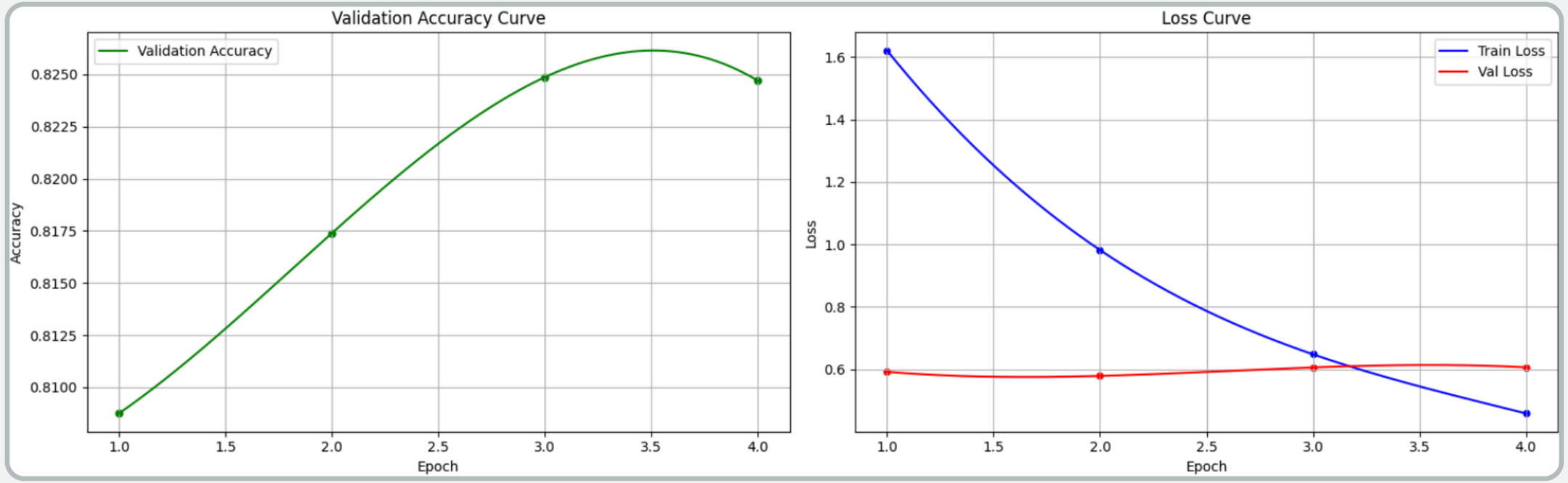
Other Parameters

- Max sequence length: 128
- Early stopping: Enabled
- Gradient accumulation: 2

BERT RESULTS

Tag	Precision	Recall	F1-score	Support
Not Bully	0.8725	0.84	0.856	5401
Threat	0.6904	0.8145	0.7473	345
Religious	0.9227	0.9172	0.9199	1522
Troll	0.7169	0.7488	0.7325	2655
Sexual	0.805	0.8164	0.8107	2119
Accuracy			0.8248	12042
Macro avg	0.8015	0.8274	0.8133	12042
Weighted avg	0.8274	0.8248	0.8257	12042

ACCURACY & LOSS CURVE



XLM-R MODEL PARAMETERS

Training Setup

- Learning rate: 2e-5
- Batch size: 16
- Epochs: 6

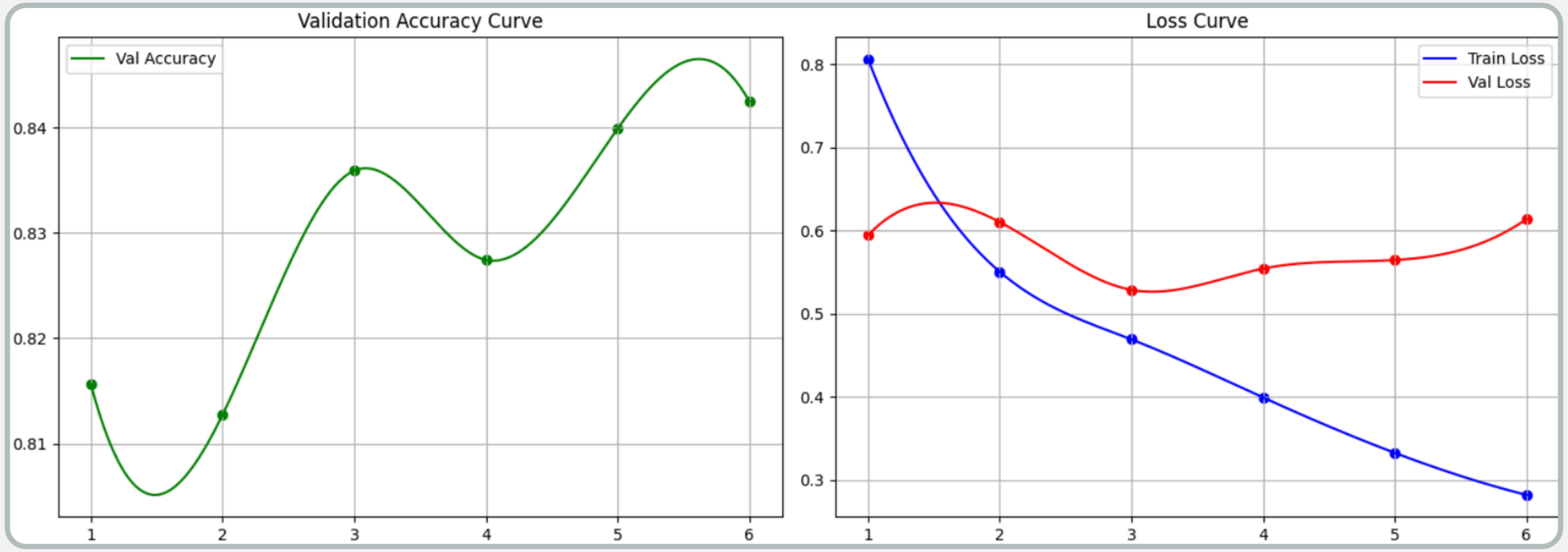
Other Parameters

- Max sequence length: 256
- Weight decay: 0.01
- Class-weighted loss used

XLM-R RESULTS

Tag	Precision	Recall	F1-score	Support
Not Bully	0.8918	0.8473	0.869	5401
Threat	0.7117	0.8087	0.7571	345
Religious	0.9088	0.9297	0.9191	1522
Troll	0.7527	0.7748	0.7635	2655
Sexual	0.8161	0.8584	0.8367	2119
Accuracy			0.8426	12042
Macro avg	0.8162	0.8438	0.8291	12042
Weighted avg	0.8448	0.8426	0.8432	12042

ACCURACY & LOSS CURVE



ENSEMBLE PARAMETERS

Models Used

- BERT
- XLM-RoBERTa

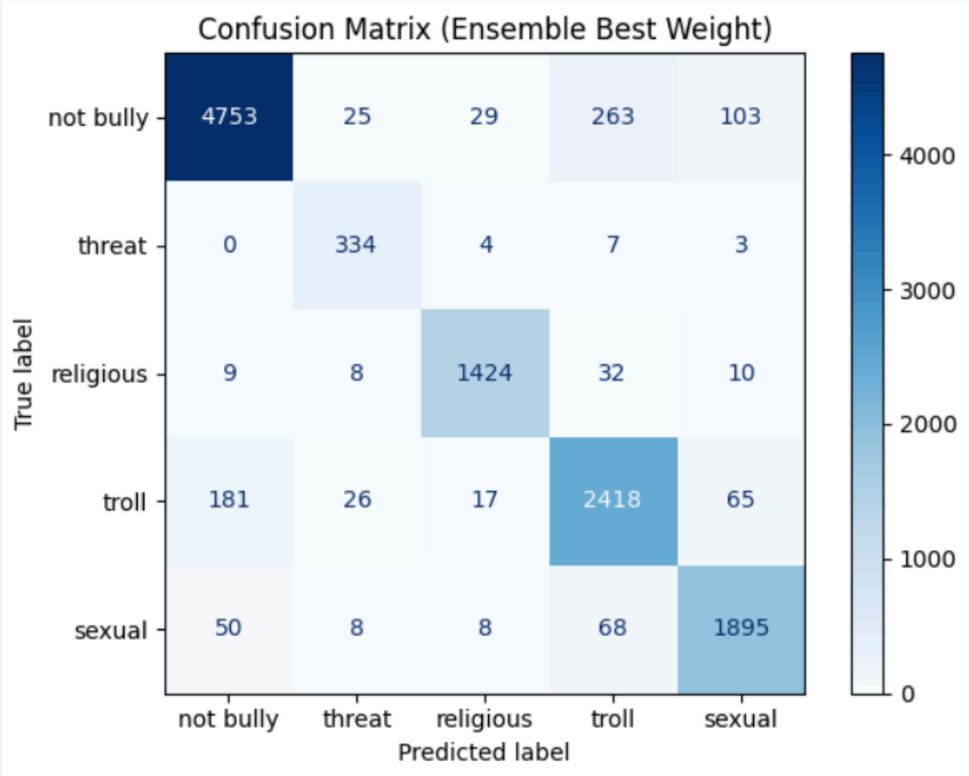
Technique

- Soft Voting Ensemble

ENSEMBLE RESULTS

Tag	Precision	Recall	F1-score	Support
Not Bully	0.9519	0.9188	0.9351	5173
Threat	0.8329	0.9598	0.8919	348
Religious	0.9609	0.9602	0.9605	1483
Troll	0.8673	0.8932	0.8801	2707
Sexual	0.9128	0.934	0.9233	2029
Accuracy			0.922	11740
Macro avg	0.9052	0.9332	0.9182	11740
Weighted avg	0.9233	0.922	0.9223	11740

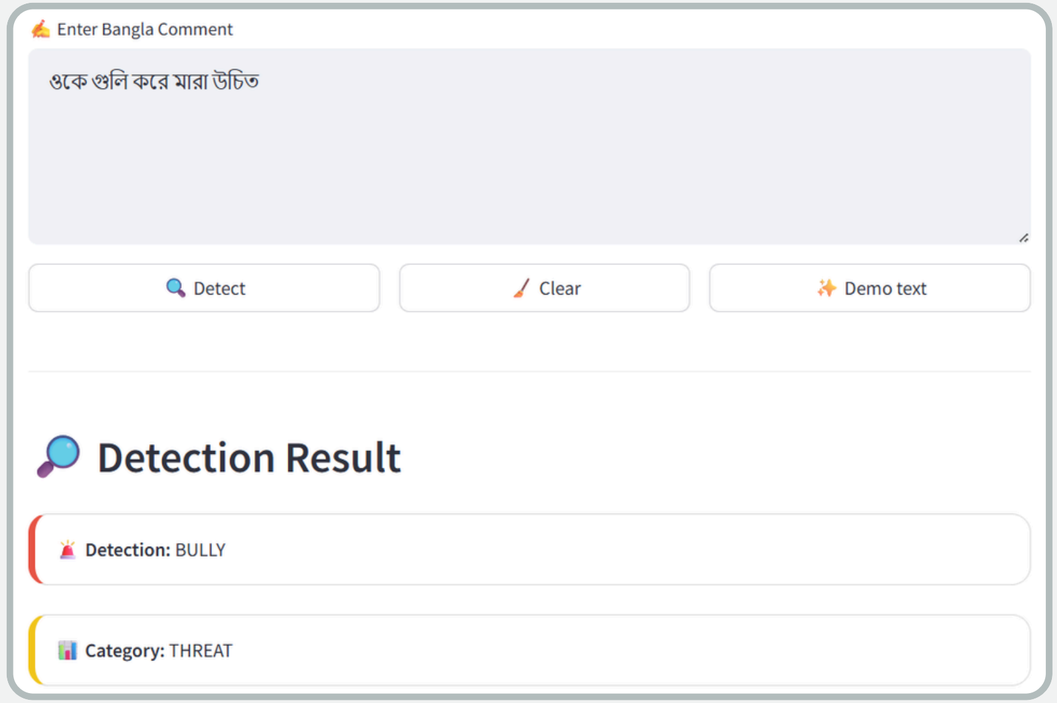
CONFUSION MATRIX

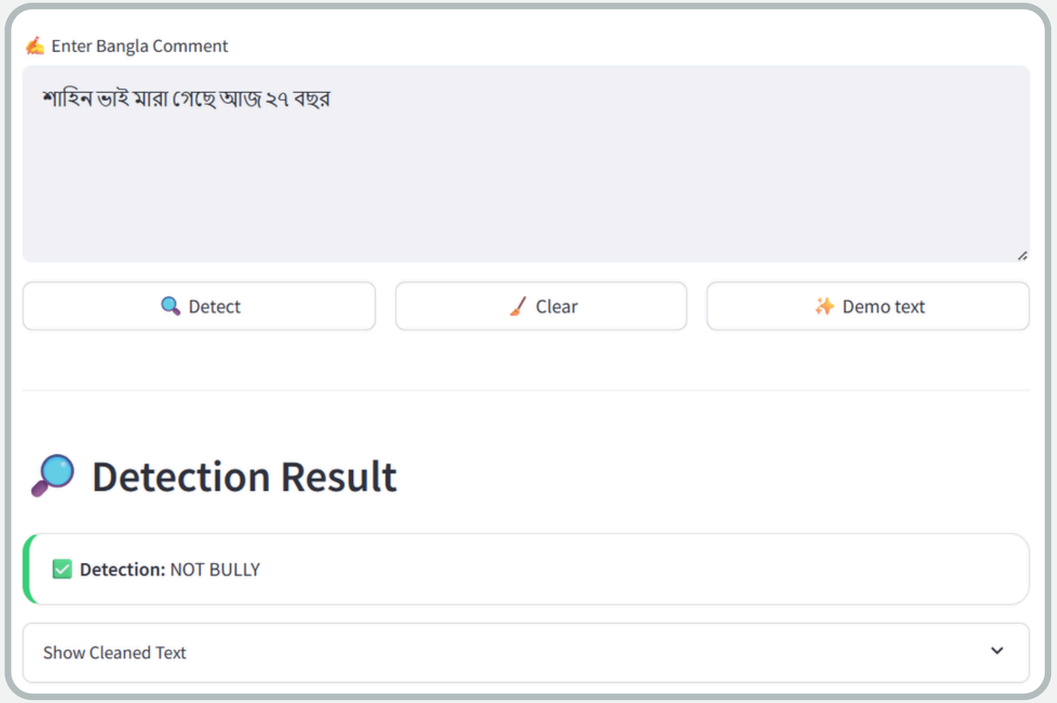


MODEL COMPARISON

Model	Accuracy	Precision	Recall	Macro F1-score
Support Vector Machine	75.94%	75.89%	75.04%	75.27%
Logistic Regression	57.44%	50.60%	51.25%	50.82%
BERT	82.47%	82.74%	82.47%	81.32%
XLM-R	84.25%	84.47%	84.25%	84.31%
Ensemble	92.20%	90.52%	93.32%	91.82%

REAL-TIME DEMONSTRATION





CONCLUSION

- Developed a multi-class Bangla cyberbullying detection system.
- Compared classical ML and transformer-based models.
- Transformer models outperformed classical approaches.
- XLM-R showed strong performance on minority classes.
- Ensemble model achieved the best results (92.20% accuracy).
- Implemented a real-time detection interface using Streamlit.

FUTURE WORK

- Expand the Bangla cyberbullying dataset.
- Improve detection of rare classes such as Threat.
- Explore more advanced transformer models.
- Deploy the system for real-world social media moderation.

THANK YOU

FOR YOUR TIME AND PATIENCE